

Value Extinction Determination Module (VXDM)

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Canonical Definition

Value Extinction Determination Module (VXDM) is the third Core Module of the Value Extinction Standard (VXSS) within VFM® — Value Flow Mechanisms Architecture.

VXDM governs the formal determination of whether previously existing value has become extinct as the same relevant value within its defined lifecycle, Value Context, value dimension, and legitimate restoration boundary.

The Module receives the governed extinction-threshold result from VXTM — Value Extinction Threshold Module and determines whether the available basis is sufficient to recognize that legitimate recoverability of the relevant value has ended.

Its operational purpose is:

to establish whether the relevant value has legitimately crossed from terminal non-recoverability into formally determined Value Extinction.

VXDM does not determine whether an object physically exists, whether an asset remains legally owned, whether a technology remains technically present, or whether other forms of value continue to exist.

It determines one specific lifecycle question:

Has the same governed value ceased to remain legitimately recoverable within the defined value lifecycle?

Architectural Position

VXDM occupies the formal determination point between:

Value Extinction Threshold

and

Value Finality.

Its architectural position is:

VXCM — Value Extinction Conditions Module → VXTM — Value Extinction Threshold Module → VXDM — Value Extinction Determination Module → VXFM — Value Finality Module

VXCM establishes the relevant extinction conditions.

VXTM establishes whether the governed extinction threshold has been reached.

VXDM determines whether Value Extinction is formally established.

VXFM governs the resulting lifecycle finality.

Therefore:

Condition ≠ Threshold.

Threshold ≠ Determination.

Determination ≠ Finality.

Internal Flow

Governed Extinction Threshold Result → Governed Value Confirmation → Prior Value Existence Confirmation → Extinction Basis Review → Recoverability Status Review → Restoration Boundary Review → Residual Value Test → Same Value Test → Extinction Uncertainty Assessment → Extinction Determination → Determination Status → Determination Record → VXFM — Value Finality Module

Operational Space

VXDM governs the formal determination space in which terminal non-recoverability is converted into a governed lifecycle conclusion.

Its operational space includes:

confirmation of previously existing value; confirmation of the governed value boundary; review of extinction conditions; review of extinction-threshold satisfaction; recoverability assessment; restoration-boundary confirmation; residual-value separation; dimensional extinction; contextual extinction;

partial extinction; same-value continuity; uncertainty; determination status; determination correction; determination provenance; transition toward Value Finality.

VXDM does not create extinction.

It governs the determination that extinction has occurred.

Prior Value Requirement

Value Extinction requires previously existing value.

If the relevant value never legitimately existed or emerged, it cannot later become extinct.

Therefore:

Value Never Emerged ≠ Value Extinction.

A failed expectation of future value is not the extinction of previously existing value.

A forecast value that never materialized is not extinct value.

A proposed value that was never established is not extinct value.

VXDM must therefore confirm the prior existence of the relevant value before extinction can be determined.

Extinction Determination

A Value Extinction Determination is the governed conclusion that previously existing value no longer remains legitimately recoverable as the same relevant value within its defined lifecycle and restoration boundary.

The determination must remain bounded to:

the relevant value; the relevant Value Object or relationship; the relevant value dimension; the applicable Value Context; the defined lifecycle; the restoration boundary; the applicable extinction threshold.

The determination must not automatically extend beyond those boundaries.

Threshold Reached ≠ Extinction Determined

VXTM may establish that the applicable extinction threshold has been reached.

That result authorizes formal determination.

It does not replace it.

VXDM must independently establish whether the governed basis supports the extinction conclusion.

Therefore:

Extinction Threshold Reached ≠ Value Extinction Determined.

Zero Value ≠ Extinction Determination

A value may remain at zero while retaining restoration potential.

Accordingly:

ZVA™ ≠ Extinct Value.

The existence of a zero-value state is insufficient to determine extinction.

VXDM must establish terminal non-recoverability of the same governed value.

Nullification ≠ Extinction Determination

Value Nullification governs the transition through which value reaches zero.

Value Extinction governs the terminal lifecycle condition in which recovery of that same value is no longer legitimate within the governed lifecycle.

Therefore:

Value Nullified ≠ Value Extinct.

A value may remain nullified for a significant period and later undergo ReMonet™ if restoration potential remains.

Failed Restoration ≠ Extinction

One unsuccessful restoration attempt does not establish extinction.

Several unsuccessful attempts do not automatically establish extinction either.

VXDM must distinguish:

Restoration Attempt Failed

from

Legitimate Restoration Pathway No Longer Exists.

Therefore:

Failed ReMonet™ ≠ Value Extinction.

Extinction concerns terminal recoverability, not the success or failure of one intervention.

Recoverability Confirmation

Before determining extinction, VXDM must confirm that the relevant recoverability boundary has been legitimately exhausted.

This may require consideration of:

remaining restoration pathways; residual technical recoverability; functional recoverability; contextual recoverability; relational recoverability; informational recoverability; temporal recoverability; dependency reconstruction;

legitimate alternative restoration mechanisms; unresolved uncertainty.

Where material recoverability remains:

Value Extinction should not be determined.

Same Value Test

VXDM must determine whether a possible future recovery pathway would restore:

the same relevant value

or create:

new or replacement value.

This distinction prevents the indefinite preservation of an extinguished lifecycle merely because another object, technology, process, or relationship may later produce a similar benefit.

Therefore:

Functional Replacement ≠ Restoration of the Same Value.

Replacement Value ≠ Restored Value.

Restoration Boundary Test

A potential pathway must remain within the legitimate restoration boundary to prevent extinction.

If recovery would require creation of a fundamentally new value relationship, the existence of that possibility does not necessarily preserve the original value from extinction.

The key question is not:

Can value exist here again?

The key question is:

Can this same governed value legitimately be recovered?

Residual Value Test

The extinction of one value does not imply extinction of every value associated with the same object.

VXDM must separate the relevant value dimension from residual value.

An industrial machine may lose its productive value permanently while retaining:

material value; component value; informational value; historical value; research value; recycling value.

Therefore:

Extinction of One Value Dimension $\neq$ Extinction of the Value Object.

Value Extinction $\neq$ Object Extinction

An object may continue to exist after a relevant value becomes extinct.

Likewise, physical destruction of an object does not automatically prove that every value associated with it has become extinct.

For example, destruction may eliminate operational value while historical, informational, evidential, intellectual, or material value persists elsewhere.

Therefore:

Object Destruction $\neq$ Automatic Value Extinction.

Value Extinction $\neq$ Object Extinction.

Dimensional Extinction

VXDM may determine extinction for a specific value dimension.

For example:

Operational Value $\rightarrow$ Extinct

while:

Material Value $\rightarrow$ Existing

and:

Historical Value $\rightarrow$ Existing.

This allows Value Extinction to remain precise rather than object-wide by default.

Partial Extinction

A governed value may consist of multiple separable components.

Where only part becomes terminally unrecoverable, VXDM may determine:

Partial Value Extinction.

The remaining value continues through its applicable lifecycle.

Therefore:

Partial Extinction $\neq$ Total Extinction.

Contextual Extinction

Value may become extinct within one context while remaining valid within another.

VXDM must therefore preserve the Value Context attached to the determination.

For example, a capability may permanently cease to have value in one operational environment while retaining value in another.

Therefore:

Contextual Value Extinction $\neq$ Universal Value Extinction.

Temporal Extinction

Some value exists only within a specific temporal relationship.

Once that relationship becomes irreversibly unavailable, the relevant time-dependent value may become extinct.

VXDM may determine extinction of that specific temporal value without concluding that all future value associated with the object has ended.

Therefore:

Expired Temporal Value $\neq$ Extinction of All Future Value.

Economic Non-Viability

An asset may cost more to restore than the expected economic return.

This may justify abandonment.

It does not automatically establish Value Extinction.

VXDM must distinguish:

Not Economically Worth Restoring

from

No Longer Legitimately Recoverable as the Same Value.

Therefore:

Abandonment Decision $\neq$ Value Extinction Determination.

Organizational Decision $\neq$ Extinction

An organization may choose to:

discontinue a product; close a facility; abandon a technology; terminate a project; dispose of an asset; stop funding restoration.

These decisions may affect value.

They do not independently determine whether the relevant value has become extinct.

Therefore:

Decision Not to Restore $\neq$ Inability to Restore.

Legal Termination $\neq$ Automatic Value Extinction

A legal right, authorization, contract, license, or permission may expire or terminate.

This may materially affect value.

However, legal termination is an external condition whose value consequence must still be governed.

VXDM determines whether the resulting loss causes the relevant value to become terminally unrecoverable.

It does not determine the legality itself.

Extinction Determination States

VXDM may assign one of the following determination states:

1. Not Extinct

Material legitimate recoverability remains.

2. Extinction Not Established

The available basis does not support a formal extinction conclusion.

3. Extinction Indeterminate

Material uncertainty prevents a reliable determination.

4. Conditionally Extinct

The extinction determination applies only under explicitly defined conditions.

5. Partially Extinct

Only part of the governed value has become terminally unrecoverable.

6. Dimensionally Extinct

A defined value dimension is extinct while other value dimensions remain.

7. Contextually Extinct

The value is extinct within a defined Value Context but not necessarily elsewhere.

8. Extinct

The same governed value has become terminally unrecoverable within its defined lifecycle and restoration boundary.

These states prevent unsupported binary conclusions.

Extinction Indeterminate

Uncertainty must remain visible.

Where evidence does not establish either recoverability or terminal non-recoverability, VXDM should not force a conclusion.

Therefore:

Uncertainty $\neq$ Extinction.

An indeterminate determination preserves the lifecycle until sufficient governance basis exists for another conclusion.

Determination Evidence Boundary

VXDM may rely on governed evidence and validated claims concerning:

physical conditions; technical impossibility; dependency loss; information loss; contextual change; restoration attempts; legal constraints; temporal boundaries; economic conditions; operational reality.

However, VXDM does not own validation or audit of that evidence.

VALIDOS® governs validation.

ADIT® governs auditability and evidence architecture.

VXDM governs:

the value-lifecycle determination derived from governed inputs.

Determination Provenance

A Value Extinction Determination should remain traceable to the lifecycle history of the relevant value.

That history may include:

Value Origin → Value Emergence → Value Recognition → Value State → Value Change → Value Degradation → Value Nullification → Restoration Attempts

→ Extinction Conditions

→ Extinction Threshold

→ Extinction Determination

Not every value must pass through every stage.

The provenance chain should reflect the actual lifecycle.

Determination $\neq$ Historical Erasure

Once extinction is determined, the prior value does not disappear from the historical record.

Its prior existence remains part of governed value provenance.

Therefore:

Value Extinction $\neq$ Erasure of Prior Value.

This distinction becomes critical for auditability, historical analysis, accountability, and future interpretation.

Correction of an Extinction Determination

An extinction determination may later be shown to have been incorrect because:

evidence was incomplete; a recovery pathway already existed but was unknown; the value boundary was incorrectly defined; the restoration boundary was too narrow; a dependency was incorrectly believed to be permanently lost; the threshold was incorrectly applied.

In such a case, the architecture should treat the issue as:

Correction of an Extinction Determination

rather than automatically:

Restoration of Extinct Value.

This preserves methodological integrity.

New Technology After Extinction

A future technology may create new capability around an object whose prior value was legitimately determined extinct.

VXDM must distinguish two possibilities.

Case A — Prior Determination Was Incorrect

The value was actually recoverable at the time of determination.

The appropriate response is correction or reassessment.

Case B — New Conditions Create a New Value Relationship

The prior value was legitimately extinct under its governed lifecycle, but genuinely new conditions later produce new value.

The appropriate pathway may be:

Value Emergence

rather than:

Value Restoration.

Therefore:

Later New Value $\neq$ Automatic Reversal of Value Extinction.

Extinction and ReMonet™

Before extinction:

Degraded / Dormant / Zero Value $\rightarrow$ ReMonet™ $\rightarrow$ Restored Value

After legitimate Value Extinction:

the ReMonet™ pathway for that same extinguished value within that lifecycle is closed.

If value later appears through a genuinely new relationship:

New Value Emergence

may begin a new governed lifecycle.

Boundary with VXTM

VXTM determines:

Has the extinction threshold been reached?

VXDM determines:

Does the governed basis establish that the relevant value is extinct?

Therefore:

Threshold Status $\neq$ Extinction Status.

Boundary with VXFM

VXDM determines extinction.

VXFM governs the lifecycle consequences and finality resulting from that determination.

Therefore:

Extinction Determination $\neq$ Value Finality.

Boundary with VRSS

VRSS governs recovery of previously existing value while legitimate restoration potential remains.

VXDM governs the determination that this restoration pathway has terminally ended for the same relevant value.

Therefore:

Restoration Potential Retained $\rightarrow$ VRSS

Restoration Potential Terminally Ended $\rightarrow$ VXDM Extinction Determination

Boundary with VNS

VNS governs Value Nullification and transition to zero.

VXDM governs whether that zero-value condition has become terminal.

Therefore:

ZVA™ may exist without Value Extinction.

Minimum Implementation Framework

A minimum VXDM implementation should establish:

1. Governed Value Confirmation

Identify the exact value under determination.

2. Prior Value Confirmation

Confirm that the relevant value previously existed.

3. Threshold Result

Receive and verify the governed VXTM threshold result.

4. Extinction Basis

Establish the conditions supporting terminal non-recoverability.

5. Recoverability Review

Confirm that material legitimate restoration pathways do not remain.

6. Same Value Test

Distinguish restoration of the same value from replacement or New Value Emergence.

7. Residual Value Test

Identify value dimensions that remain outside the extinction determination.

8. Uncertainty Assessment

Preserve unresolved uncertainty where material.

9. Extinction Determination

Assign the appropriate determination status.

10. Finality Transfer

Transfer the governed determination to VXFM for Value Finality governance.

Governance Outputs

VXDM may produce:

Value Extinction Determination Extinction Determination Status Governed Value Boundary Record Prior Value Confirmation Recoverability Determination Basis Restoration Boundary Confirmation Residual Value Assessment Same Value Assessment Partial Extinction Determination Dimensional Extinction Determination Contextual Extinction Determination Extinction Uncertainty Record Extinction Determination Provenance Finality Transfer Record

Enterprise Application

VXDM gives organizations a formal governance mechanism for determining when previously existing value should cease to be treated as recoverable within the same lifecycle.

It can support governance of:

obsolete industrial assets; stranded infrastructure; discontinued technologies; abandoned intellectual property; inaccessible digital assets; permanently lost datasets; expired capabilities; failed products; legacy systems; distressed investments; organizational capabilities; long-term restoration programs.

Its principal enterprise function is to prevent both:

premature write-off of recoverable value

and

indefinite preservation of value that has legitimately reached terminal non-recoverability.

Use Case 1 — Extinction of Industrial Productive

Value

A specialized production system previously generated substantial productive value.

Over time:

Positive Productive Value → Degradation → DeMonet™ → ZVA™

The physical machine remains.

Multiple restoration pathways are assessed.

Critical proprietary components are permanently unavailable, control architecture cannot be reconstructed, original technical dependencies have disappeared, and any modern replacement would constitute a fundamentally new production system.

VXTM establishes:

Productive Value Extinction Threshold → Reached

VXDM then evaluates:

Prior Productive Value Confirmed → Threshold Result Confirmed → Recoverability Exhausted → Same Value Cannot Be Restored → Residual Material Value Identified → Productive Value Extinction Determination

Result:

Original Productive Value → Extinct

while:

Material Value → Persists

and potentially:

Historical / Informational Value → Persists

The object survives.

The specific productive value does not.

Use Case 2 — Irrecoverable Historical Dataset

An organization possesses a historical dataset whose analytical value depended on unique underlying records.

The records become inaccessible following irreversible storage destruction.

No backup exists.

No secondary source can reconstruct the original dataset with sufficient continuity.

Partial summaries remain, but they cannot restore the original informational value.

VXTM establishes that the extinction threshold for the original dataset's governed informational value has been reached.

VXDM evaluates:

Prior Informational Value Confirmed → Original Data Permanently Unrecoverable → No Legitimate Reconstruction Pathway → Partial Summaries ≠ Original Dataset → Residual Derived Information Separated

→ Extinction Determination

Result:

Original Dataset Informational Value → Extinct

The surviving summaries may retain their own value.

A future dataset created from new sources may generate:

New Value Emergence

but does not automatically restore the extinguished informational value of the original dataset.

Foundational Principles

Value Never Emerged $\neq$ Value Extinction.

Zero Value $\neq$ Value Extinction.

ZVA™ $\neq$ Extinct Value.

Value Nullification $\neq$ Value Extinction.

Failed Restoration $\neq$ Value Extinction.

Threshold Reached $\neq$ Extinction Determined.

Uncertainty $\neq$ Extinction.

Decision Not to Restore $\neq$ Inability to Restore.

Abandonment $\neq$ Value Extinction.

Economic Non-Viability $\neq$ Automatic Value Extinction.

Object Destruction $\neq$ Automatic Extinction of Every Value Dimension.

Value Extinction $\neq$ Object Extinction.

Partial Extinction $\neq$ Total Extinction.

Contextual Extinction $\neq$ Universal Extinction.

Replacement Value $\neq$ Restored Value.

Functional Replacement $\neq$ Restoration of the Same Value.

Extinction Determination $\neq$ Value Finality.

Value Extinction $\neq$ Erasure of Prior Value.

Later New Value $\neq$ Automatic Reversal of Value Extinction.

A legitimate Value Extinction determination closes restoration of the same governed value within that lifecycle, not the possibility that new value may emerge in the future.

Canonical Closing Statement

Value Extinction Determination Module establishes the formal governance point at which previously existing value may be recognized as extinct within VFM®. By requiring confirmation of prior value, threshold satisfaction, terminal non-recoverability, restoration-boundary exhaustion, same-value continuity, residual value, contextual and dimensional boundaries, and material uncertainty, VXDM prevents zero value, abandonment, failed restoration, physical destruction, economic non-viability, or organizational decision from being incorrectly treated as Value Extinction. The Module preserves the distinction between extinction of value and extinction of the underlying object, maintains the provenance of previously existing value, and separates legitimate extinction from later New Value Emergence. Value becomes formally extinct only when the governed basis establishes that recovery of that same relevant value has legitimately ended within its defined lifecycle and restoration boundary.

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Governed Links

→ Value Extinction Standard (VXSS).

→ VFM® Architecture Map

→ Previous: Value Extinction Threshold Module (VXTM).

→ Next: Value Finality Module (VXFM).

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